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1. Executive Summary

As a research company committed to delivering rigorous and insight-driven solutions, our objective is to support Apple in addressing key strategic challenges within an increasingly competitive and dynamic global technology market. This research proposal is designed to generate actionable insights that inform decision-making across product innovation, pricing strategy, ecosystem development, and sustainability positioning.

This study aims to support Apple across four strategic areas. First, to understand how product innovation and diversified service offerings contribute to market expansion across key regions. Second, to evaluate how economic conditions shape consumers' purchasing decisions and influence Apple's pricing effectiveness. Third, to assess how privacy protection and ecosystem integration drive customer trust, loyalty, and device retention, particularly among Generation Z and millennial consumers. Lastly, to examine how Apple's sustainability initiatives influence brand perception and long-term loyalty across geographic markets.

Guided by clearly defined Research Aims (RAs), our research team will systematically formulate Research Problems (RPs) and Research Objectives (ROs) to ensure alignment throughout the study. This structured framework ensures that all survey instruments, analytical techniques, and insights remain tightly focused on Apple's strategic priorities.

Survey instruments will be developed through a rigorous research process incorporating secondary research, exploratory focus groups, and pilot testing to ensure validity and reliability. Probability-based sampling methods will be employed to obtain a representative dataset, supported by supplementary approaches to enhance sample robustness and statistical inference.

The collected data will be analysed using a suite of advanced statistical techniques aligned with the research objectives, including factor analysis, cluster analysis, one-way ANOVA, multiple linear regression, and independent-samples t-tests. These analyses will generate actionable insights into consumer behaviour, preference segmentation, and perceptual differences across groups. Together, the findings will enable Apple to strengthen its competitive positioning, enhance customer retention, and optimise sustainability initiatives for long-term strategic impact.

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2. Background

Established in 1976, Apple Inc. is a leading global technology company recognised for its advancements in consumer hardware, software solutions, and digital services, particularly in the premium smartphone, wearable, and ecosystem-based markets. Despite intense competition from brands such as Samsung, Google, Microsoft and other Android manufacturers, Apple continues to maintain a strong market-leading position.

In response to growing competition and market saturation, Apple continues to expand its ecosystem and services while reinforcing its commitments to sustainability, privacy, and social responsibility. As CEO Tim Cook emphasises, *“We believe that business, at its best, serves the public good, empowers people around the world, and binds us together as never before.”* Apple’s core philosophy underpins Apple’s broader corporate values and guides how it approaches innovation, customer experience, and ethical business practices across markets.

3. Problem Definition

3.1 Introduction

Driven by post-pandemic digitalisation, remote work, and growing demand for premium, privacy-focused technology, Apple continues to experience strong adoption across key markets such as the United States and Asia-Pacific. This growth is reflected in Apple’s total revenue, which increased from approximately US\$260.2 billion in 2019 to US\$416 billion in 2025 (StockAnalysis, 2026).

To sustain its leadership, Apple emphasises product innovation, ecosystem integration, and service expansion while delivering seamless, personalised user experiences. Understanding consumer preferences related to innovation, pricing, privacy protection, and sustainability is therefore critical, particularly among Gen Z and millennials.

This research aims to provide insights into consumer preferences and satisfaction across Apple’s products and services, supporting data-driven strategies to strengthen competitive advantage while upholding Apple’s social and environmental commitments.

3.2 Research Aims (RAs)

Based on the client brief, this report outlines four key core Research Aims (RAs) aligned with Apple’s strategic and research priorities.

RA1: To examine how product innovation and diversified services support Apple’s global market expansion across key regions.

RA2: To assess how economic conditions influence consumer purchasing behaviour and the effectiveness of adaptive pricing strategies.

RA3: To evaluate how privacy protection and ecosystem integration affect customer trust, loyalty, and device retention, particularly among Generation Z and millennials.

RA4: To analyse how Apple's sustainability initiatives can be effectively communicated to strengthen brand equity and long-term customer loyalty.

3.3 Research Problems (RPs)

Guided by the defined Research Problems (RPs), the Research Objectives (ROs) are developed to ensure that the study delivers precise, relevant, and actionable insights for Apple.

RP1 (for RA1): Examine how consumers evaluate Apple's existing product lines and service offerings in order to identify which aspects of product innovation and ecosystem services most strongly support market expansion across regions.

RP2 (for RA2): Investigate how key economic factors—such as employment and income levels, inflation, interest rate and consumer confidence—influence consumers' purchase intentions for Apple products (Investopedia., 2025).

RP3 (for RA3): Assess the extent to which Apple's privacy protection and ecosystem integration influence device retention and long-term loyalty among Generation Z and millennial consumers.

RP4 (for RA4): Examine how consumers perceive Apple's sustainability initiatives and determine which environmental dimensions most strongly shape brand perception across different geographic markets.

4. Research Objectives (ROs)

Our Research Problems (RPs) inform the development of Research Objectives (ROs), ensuring that the study offers Apple precise and effective insights.

RA1

RO1: A structured consumer survey will be conducted across key geographic regions, where respondents will rate their satisfaction with Apple's major product and service categories using a 7-point Likert scale. Mean satisfaction scores will be compared across regions to identify offerings that contribute most strongly to Apple's market position.

Respondents will also rate the importance of key product and service attributes, including performance, design, ecosystem integration, and privacy features. Cluster analysis will be applied to these ratings—separately for Apple users and for a combined Apple and competitor sample—to identify distinct consumer segments and assess how innovation and service offerings support global market expansion.

RA2

RO2: Key economic factors influencing consumers' purchasing decisions will first be identified through interviews and operationalised into survey items measured on a 7-point Likert scale. These factors include perceived price levels, financing or instalment costs, personal income and job stability, and overall economic uncertainty.

Participants will also report their likelihood of purchasing an Apple product within the next 12 months using a continuous 0–100 scale, which serves as the dependent measure of purchase intention.

Multiple linear regression will then be applied to evaluate the relative influence of each economic factor on purchase intention. This analysis enables Apple to identify which economic conditions most strongly shape purchasing behaviour and to refine pricing, financing, and trade-in strategies accordingly. For example, a one-point increase in perceived price sensitivity may be associated with a decrease in the predicted likelihood of purchasing an Apple product.

RA3

RO3.1: In the survey, participants will evaluate the perceived importance and effectiveness of Apple's privacy and security features using a 7-point Likert scale.

Participants will also indicate their likelihood of continuing to use Apple devices due to privacy protection and ecosystem integration on a continuous 0–100 scale, which serves as the measure of device retention.

Factor analysis will be applied to the privacy and ecosystem features to identify underlying perception dimensions. Privacy-related features are expected to load onto a "Privacy Protection" factor, while cross-device continuity features are expected to load onto an "Ecosystem Integration" factor. Factor scores will then be examined in relation to device retention to assess which dimensions are most strongly associated with continued usage.

RO3.2: To examine whether retention drivers differ across demographic groups, independent samples t-tests will be conducted to compare average device retention scores between Generation Z and Millennials. Statistically significant differences will indicate whether retention is driven by distinct priorities across generations, informing more targeted privacy communication and ecosystem

development strategies.

RA4

RO4.1: Respondents will rate the impact of initiatives aligned with the Reduce, Reuse, and Recycle framework using a 7-point Likert scale.

Factor analysis will be conducted to group these initiatives into broader sustainability dimensions reflecting the Reduce, Reuse, and Recycle framework, simplifying interpretation and highlighting the sustainability themes most salient to consumers.

RO4.2: To assess whether sustainability perceptions differ across consumer groups, one-way ANOVA will be used to compare average brand perception scores across demographic segments such as age groups or regions. Statistically significant differences will indicate where sustainability communication and initiatives can be more effectively tailored to strengthen brand equity and global positioning.

5. Research Design

We propose a two-stage research design comprising exploratory and descriptive phases to generate insights aligned with the Research Objectives (ROs).

5.1 Exploratory Phase

We propose conducting online focus groups with five participants per group, each lasting approximately 1.5 hours, across Apple's major sales regions (Americas, Japan, Europe, Greater China, and Rest of Asia-Pacific). Participants will be recruited from high-sales areas identified through Apple's internal database. Online qualitative interviews are chosen for their global reach, cost efficiency, and scheduling flexibility, with trained local moderators engaged where necessary to minimise language barriers and response bias.

In the exploratory phase, we will gather insights on consumer satisfaction, product attribute preferences, pricing perceptions, privacy attitudes, and sustainability awareness across Apple's products and services. Relevant secondary data on sustainability initiatives will also be collected and validated with Apple. These insights will be incorporated into the survey for the subsequent descriptive phase.

5.2 Descriptive Phase

Building on exploratory insights, a cross-sectional survey will be implemented using structured questionnaires with Likert and semantic differential scales to measure consumer perceptions and attitudes. A two-stage data collection approach is recommended: an initial email survey followed by a

postal mail survey to non-respondents after two weeks. This approach aims to reduce non-response bias and accommodate participants with limited internet access. Incentives will be used to improve response rates, and the questionnaire will be designed to maximise clarity and respondent convenience.

Prior to full deployment, a pilot study will be conducted to test questionnaire design, scale structure, translations, and survey flow. Validity and reliability will be assessed using appropriate techniques, including test-retest reliability where feasible. This ensures the completed survey provides optimal representation of the intended data and feedback. The pilot will initially involve approximately 15–20 participants per region to allow close observation of respondent behaviour, before transitioning online for cost efficiency. Pilot testing will continue iteratively until no further revisions are required, ensuring alignment with the research objectives.

5.3 Statistical Tools

Every survey item is intentionally crafted to identify and shape variables that contribute to achieving our Research Objectives (ROs), under the guidance of our Research Aims (RAs) in responding to the Research Problems (RPs).

Relevant analytical approaches will be used for every question, with hypothesis testing carried out 5% significance level ($\alpha = 0.05$).

RA1

- 1) RO1 - Cluster Analysis

RA2

- 2) RO2 - Multiple Linear Regression

RA3

- 3) RO3.1 - Factor analysis
- 4) RO3.2 - Independent T-test

RA4

- 5) RO4.1 - Factor Analysis
- 6) RO4.2 – ANOVA

6. Data Collection/Fieldwork

Utilising Apple’s customer database and our research panel, the target population will include existing Apple customers as well as non-Apple users who currently use competitor brands. This enables for comparative analysis between current and potential customers.

Our sampling frame will consist of customer email and mailing addresses obtained through Apple ID records and our access panel. A stratified random sampling approach will be employed, with respondents stratified by Apple’s five global sales regions in proportion to their regional market representation. Within each stratum, simple random sampling will be applied. This approach ensures regional representativeness, reduces sampling bias, and aligns with Apple’s global market structure.

As a precaution to achieve sufficient participation, we suggest implementing snowball sampling. The survey link, accessible via QR code, will be shared through Apple’s online platforms, and participants will be encouraged to forward it to others in exchange for added incentives. This approach strengthens representativeness and boosts sample size, facilitating valid statistical inference.

To determine a sufficient initial sample size, both eligibility (incidence rate) and expected survey completion rate are incorporated using the following formula:

$$\text{Initial Sample Size} = \frac{\text{Required Final Sample Size}}{\text{Incidence rate} \times \text{Completion rate}}$$

The incidence rate indicates the percentage of individuals who qualify to take part in the study. Completion rate, on the other hand, is the proportion of those eligible individuals who complete the interview, enabling researchers to consider potential refusals.

According to NiCE benchmarks, the average response rate for emailed B2C relationship surveys is approximately 13% (NICE, n.d.). As surveys distributed via Apple’s database target verified Apple users, the incidence rate for Apple customers is assumed to be 100%. To achieve a minimum of 5,000 completed responses from Apple customers, approximately 38,500 invitations will be required.

$$\text{Initial Sample Size} = \frac{5,000}{1.00 \times 0.13} = 38,462$$

In addition, 5,000 non-Apple users will be sampled to enable comparative analysis between customers and non-customers. This comparison will provide insights into unmet needs, adoption barriers, and potential drivers of market expansion. Given that Apple’s market share is 27.93%, the incidence rate for non-customers is 100%-27.93%=72.07%. The omnibus survey response rate is 65% (Thomas, K., & Coleman, P., 2004). To achieve a minimum of 5,000 completed responses from Non-Apple customers, an initial invitation sample of approximately 10,700 respondents will be required.

$$\text{Initial Sample Size} = \frac{5,000}{0.7207 \times 0.65} = \frac{5,000}{0.468455} = 10,672.73$$

All samples will be segmented across the five global sales regions in proportion to Apple’s regional market share, based on Apple’s 2025 financial disclosures.

7. Data Analysis

7.1 Research Analysis 1 (for RA1)

Mean satisfaction scores and associated 95% confidence intervals will be calculated for each product category and compared across key geographic regions.

Cluster analysis will be conducted on data derived from two key attribute-importance questions: one completed by Apple users only and another by both Apple and competitor users. The Apple-only clustering captures attribute salience within Apple's existing customer base, while the Apple-and-competitor clustering identifies broader market-level preference structures relevant to potential switching behaviour.

Reliability and Brand Reputation
Performance and Specifications
Compatibaility and Ecosystem
Build Quality and Design
Security and Privacy

Figure 8.1 - For competitor users

Apple Silicon performance (A-series/M-series chips)
ProMotion display (120Hz smoothness)
Advanced camera features (Night mode, Photonic Engine, ProRAW)
Dynamic Island (only for iPhone)
Apple's cross-device ecosystem (AirDrop, Handoff, Universal Clipboard)
Face ID and Secure Enclave
App Tracking Transparency
MagSafe wireless charging system
Live Text and on-device intelligent features
iCloud syncing and continuity features

Figure 8.2 - For Apple users

Three different measures will be applied, as presented below, enabling comparative analysis, outlier detection, and similarity measurement.

$$\text{Euclidean distance: } S_{ij} = \sqrt{\sum_{k=1}^p (x_{ik} - x_{jk})^2} \quad \text{Manhattan distance: } S_{ij} = \sum_{k=1}^p |x_{ik} - x_{jk}| \quad \text{Chebyshev distance: } S_{ij} = \max_{k=1, \dots, p} (|x_{ik} - x_{jk}|)$$

Following an assessment of pairwise dissimilarities via the proximity matrix, Ward's minimum variance hierarchical agglomerative clustering will be employed for both analyses. The dendrogram and agglomeration schedule will be used to determine the optimal cluster solution, after which clusters will be profiled using geodemographic variables (region, age) to support interpretation of consumer preference heterogeneity.

7.2 Research Analysis 2 (for RA2)

Multiple linear regression will be employed for RO2.0, with key economic factors influencing Apple purchase decisions specified as independent variables and consumers' purchase intention as the dependent variable. Purchase intention will be measured on a continuous 0–100 scale to enhance sensitivity and better satisfy linear regression assumptions.

Rising prices make me think twice before buying new devices.	
Higher installments or financing costs make these devices feel less affordable to me.	
I pay attention to the exchange rate changes before buying an Apple device.	
When the economy feels uncertain, I avoid buying high-priced electronics.	

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \varepsilon_i$$

Where:

- Y_i = Purchase intention for respondent i (0–100 scale)
- β_0 = Intercept
- β_j = Population regression coefficient for economic factor j
- ε_i = Error term

$$\hat{Y}_i = b_0 + b_1 X_{1i} + b_2 X_{2i} + b_3 X_{3i} + b_4 X_{4i}$$

Where:

- $\hat{Y}_i$ = Predicted purchase intention
- b_0 = Estimated intercept
- b_1, b_2, b_3, b_4 = estimated population regression coefficients

Model	Unstandardised Coefficients		Standardised Coefficient Beta	t	sig.
	B	Std. Error			
Constant					
X1					
X2					
X3					
X4					

Table 2.1

Under unstandardised coefficients (B), we assess the magnitude and direction of each economic factor's impact on purchase intention. Variables with statistically significant coefficients indicate key economic drivers influencing Apple purchasing behaviour.

Model	R	R square	Adjusted R Square	Std. Error of the Estimate
1				

Table 2.2

Adjusted R-square is used to assess the explanatory power of the regression model while accounting for model complexity and mitigating overfitting. A higher adjusted R-square indicates that the selected economic variables collectively explain a substantial proportion of the variance in consumers' purchase intention.

Model	Sum of Squares	Degree of Freedom	Mean Square	F	Sig.
Regression					
Residual					
Total					

Table 2.3

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$$

$$H_1 : \text{At least one } \beta_j \neq 0 \text{ where } j = 1, 2, 3, 4$$

The F-test is examined to assess overall model significance. A significance level of less than 0.05 results in the rejection of H_0 of zero-valued regression coefficients, suggesting that the model has explanatory power.

$$H_0 : \beta_j = 0$$

$$H_1 : \beta_j \neq 0 \text{ where } j = 1, 2, 3, 4$$

Finally, for each individual independent variable, A significance value below 0.05 results in the rejection of the H_0 , suggesting that the economic factor has a significant effect on consumers' purchase intention for Apple products.

7.3 Research Analysis 3.1 (for RA3)

Factor analysis will be employed for RO3.1. Each observed variable from Section 3 of the questionnaire is modelled as a linear combination of common factors and a unique error term.

Private Browsing: Safari keeps your activity private by not saving history, searches, or form data, blocking trackers, and locking windows when not in use.
Password Monitoring: Checks to see whether your saved Keychain passwords have been compromised in data breaches.
Location fuzzing: Apple Maps obscures your precise location by converting it to a less exact one after 24 hours and does not store your search or location history.
Third-party app permissions: Apple's photo picker lets you control exactly which photos an app can access, shows what content apps can use, and provides reminders so you can change permissions.
End-to-end encryption: iMessage and FaceTime messages are encrypted on your device so only you can access them, and Apple cannot read them in transit.
In-store merchant transactions: Payments use a secure device number and dynamic code, keeping your card and personal info private unless you give permission.

$$X_i = A_{i1}F_1 + A_{i2}F_2 + A_{i3}F_3 + \dots + A_{im}F_m + V_iU_i$$

where:

- X_i = the i th standardised variable
- A_{ij} = the standardised multiple regression coefficient of variable i on common factor j
- F = a common factor
- V_i = the standardised regression coefficient of variable i on unique factor i
- U_i = the unique factor for variable i
- m = the number of common factors.

$$F_i = W_{i1}X_1 + W_{i2}X_2 + W_{i3}X_3 + \dots + W_{ik}X_k$$

where:

- F_i = the estimate of the i th factor
- W_i = the weight or factor score coefficient
- k = the number of variables.

	Variables	Private browsing	Password monitoring	Location Fuzzing	Third-party app permissions	End-to-end Encryption	In-store Merchant Transaction
Correlation	Private browsing	1					
	Password monitoring		1				
	Location Fuzzing			1			
	Third-party app permissions				1		
	End-to-end Encryption					1	
	In-store Merchant Transaction						1
Sig. (1-tailed)	Private browsing						
	Password monitoring						
	Location Fuzzing						
	Third-party app permissions						
	End-to-end Encryption						
	In-store Merchant Transaction						

Table 3.1 – Correlation Matrix

The correlation matrix will first be examined to assess the strength, direction, and suitability of variable relationships for factor analysis.

Kaiser-Meyer-Olkin Measure of Sampling Adequacy (KMO)		
Barlett's Test of Sphericity	Approx. Chi-Square	
	Degree of Freedom	
	Significance	

Table 3.2 – KMO and Barlett's Test

The KMO statistic will be used to evaluate sampling adequacy, where a value greater than 0.5 indicates that correlations among variables can be linked to fundamental factors. Bartlett's Test of Sphericity will also be conducted, a significance value below 0.05 suggests that the correlation matrix differs from an identity matrix, confirming the suitability of factor analysis.

	Initial	Extraction
Private browsing		
Password monitoring		
Location Fuzzing		
Third-party app permissions		
End-to-end Encryption		
In-store Merchant Transaction		

Table 3.3 - Communalities

The communalities of each variable will be assessed to determine the extent to which extracted factors explain their variance, with variables exhibiting low communalities considered for removal.

Based on prior research and theoretical expectations, a two-factor solution—Privacy Protection and Ecosystem Integration—is anticipated and will be validated using eigenvalues greater than one and inspection of the scree plot.

	Component	
	1	2
Private browsing		
Password monitoring		
Location Fuzzing		
Third-party app permissions		
End-to-end Encryption		
In-store Merchant Transaction		

Table 3.4 – Rotated Component Matrix

Using Table 3.4, variables with high loadings will be grouped to interpret and label each factor.

	Component	
	1	2
Private browsing		
Password monitoring		
Location Fuzzing		
Third-party app permissions		
End-to-end Encryption		
In-store Merchant Transaction		

Table 3.5 – Component Score Correlation Matrix

$$F_i = W_{i1}X_1 + W_{i2}X_2 + W_{i3}X_3 + \cdots + W_{ik}X_k.$$

Factor scores will then be computed using the component score correlation matrix.

Reproduced Correlation	Variables	Private browsing	Password monitoring	Location Fuzzing	Third-party app permissions	End-to-end Encryption	In-store Merchant Transaction
	Private browsing						
	Password monitoring						
	Location Fuzzing						
	Third-party app permissions						
	End-to-end Encryption						
	In-store Merchant Transaction						
Residual	Private browsing						
	Password monitoring						
	Location Fuzzing						
	Third-party app permissions						
	End-to-end Encryption						
	In-store Merchant Transaction						

Table 3.6 – Reproduced Correlations

The model is expected to exhibit a low level of non-redundant residuals, reflecting good fit.

7.4 Research Analysis 3.2 (for RA3)

Independent samples t-tests will be conducted on the factor scores derived in RO3.1. Specifically, mean factor scores for Privacy Protection and Ecosystem Integration will be compared between Generation Z and Millennials.

$$H_0 : \mu_{f, \text{Gen Z}} = \mu_{f, \text{Millennials}}$$

$$H_1 : \mu_{f, \text{Gen Z}} \neq \mu_{f, \text{Millennials}}$$

Where:

- $f \in \{\text{Privacy Protection, Ecosystem Integration}\}$

T-test Hypothesis

			Levene's Test for Equality of Variance		t-test for Equality of Means					
			F	Sig.	1	Degree of Freedom	Sig. (1-tailed)	Mean difference	Std. Error Difference	95% Confidence Interval of the Difference
Gen-Z Millennials	Privacy Protection	Equal variances assumed								
		Equal Variances not assumed								
	Ecosystem Integration	Equal variances assumed								
		Equal Variances not assumed								

Table 3.7 – Independent Sample Test

$$H_0 : \sigma_{\text{Gen Z}}^2 = \sigma_{\text{Millennials}}^2$$

$$H_1 : \sigma_{\text{Gen Z}}^2 \neq \sigma_{\text{Millennials}}^2$$

Levene's Test Hypothesis

Prior to hypothesis testing, Levene's Test will be performed to assess variance homogeneity. A significance value below 0.05 results in the rejection of H_0 , indicating unequal variances; in this case, results should be read from the 'Equal variances not assumed' row.

For example, if Millennials exhibit significantly higher mean scores on the Privacy Protection factor, while Gen Z users may demonstrate stronger emphasis on Ecosystem Integration and convenience. These findings enable Apple to tailor privacy communication and ecosystem development strategies to better align with generational priorities.

7.5 Research Analysis 4.1(for RA4)

Factor analysis will be employed for RO4.1 to identify the underlying dimensions of consumer perceptions measured in Section 4 of the questionnaire. Each observed variable is represented as a linear combination of common factors plus a unique error term.

Reduce	Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.
	All Apple offices, retail stores, and data centres run on 100% renewable energy.
	Apple helps suppliers adopt renewable energy and reduce emissions.
	Apple achieved zero waste to landfill at all established final assembly sites in 2024.
Reuse	Apple operates a global Trade-In and refurbishment program to give devices a second life.
	Replacement and repair programs prolong device use (battery replacements, screen repairs).
Recycle	Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).
	Closed-loop recycling systems in manufacturing to recover materials.
	Shifting to 100% recycled aluminium enclosures on many product lines.

$$X_i = A_{i1}F_1 + A_{i2}F_2 + A_{i3}F_3 + \cdots + A_{im}F_m + V_iU_i$$

where:

- X_i = the i th standardised variable
- A_{ij} = the standardised multiple regression coefficient of variable i on common factor j
- F = a common factor
- V_i = the standardised regression coefficient of variable i on unique factor i
- U_i = the unique factor for variable i
- m = the number of common factors.

$$F_i = W_{i1}X_1 + W_{i2}X_2 + W_{i3}X_3 + \cdots + W_{ik}X_k$$

where:

- F_i = the estimate of the i th factor
- W_i = the weight or factor score coefficient
- k = the number of variables.

		Reduce				Reuse		Recycle		
		Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.	All Apple offices, retail stores, and data centres run on 100% renewable energy.	Apple helps suppliers adopt renewable energy and reduce emissions.	Apple achieved zero waste to landfill at all established final assembly sites in 2024.	Apple operates a global Trade-In and refurbishment program to give devices a second life.	Replacement and repair programs prolong device use (battery replacements, screen repairs).	Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).	Closed-loop recycling systems in manufacturing to recover materials.	Shifting to 100% recycled aluminium enclosures on many product lines.
Correlation	Reduce	Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.	1							
		All Apple offices, retail stores, and data centres run on 100% renewable energy.		1						
		Apple helps suppliers adopt renewable energy and reduce emissions.			1					
		Apple achieved zero waste to landfill at all established final assembly sites in 2024.				1				
	Reuse	Apple operates a global Trade-In and refurbishment program to give devices a second life.					1			
		Replacement and repair programs prolong device use (battery replacements, screen repairs).						1		
	Recycle	Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).							1	
		Closed-loop recycling systems in manufacturing to recover materials.								1
		Shifting to 100% recycled aluminium enclosures on many product lines.								
										1
Fig. (1) (table)	Reduce	Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.								
		All Apple offices, retail stores, and data centres run on 100% renewable energy.								
		Apple helps suppliers adopt renewable energy and reduce emissions.								
		Apple achieved zero waste to landfill at all established final assembly sites in 2024.								
	Reuse	Apple operates a global Trade-In and refurbishment program to give devices a second life.								
		Replacement and repair programs prolong device use (battery replacements, screen repairs).								
	Recycle	Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).								
		Closed-loop recycling systems in manufacturing to recover materials.								
		Shifting to 100% recycled aluminium enclosures on many product lines.								

Table 4.1 – Correlation Matrix

Kaiser-Meyer-Olkin Measure of Sampling Adequacy (KMO)		
Barlett's Test of Sphericity	Approx. Chi-Square	
	Degree of Freedom	
	Significance	

Table 4.2 – KMO and Barlett's Test

	Initial	Extraction
Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.		
All Apple offices, retail stores, and data centres run on 100% renewable energy.		
Apple helps suppliers adopt renewable energy and reduce emissions.		
Apple achieved zero waste to landfill at all established final assembly sites in 2024.		
Apple operates a global Trade-In and refurbishment program to give devices a second life.		
Replacement and repair programs prolong device use (battery replacements, screen repairs).		
Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).		
Closed-loop recycling systems in manufacturing to recover materials.		
Shifting to 100% recycled aluminium enclosures on many product lines.		

Table 4.3 - Communalities

	Component		
	1	2	3
Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.			
All Apple offices, retail stores, and data centres run on 100% renewable energy.			
Apple helps suppliers adopt renewable energy and reduce emissions.			
Apple achieved zero waste to landfill at all established final assembly sites in 2024.			
Apple operates a global Trade-In and refurbishment program to give devices a second life.			
Replacement and repair programs prolong device use (battery replacements, screen repairs).			
Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).			
Closed-loop recycling systems in manufacturing to recover materials.			
Shifting to 100% recycled aluminium enclosures on many product lines.			

Table 4.5 – Component Score Correlation Matrix

$$F_i = W_{i1}X_1 + W_{i2}X_2 + W_{i3}X_3 + \cdots + W_{ik}X_k.$$

			Reduce				Reuse		Recycle		
			Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.	All Apple offices, retail stores, and data centres run on 100% renewable energy.	Apple helps suppliers adopt renewable energy and reduce emissions.	Apple achieved zero waste to landfill at all established final assembly sites in 2024.	Apple operates a global Trade-In and refurbishment program to give devices a second life.	Replacement and repair programs prolong device use (battery replacements, screen repairs).	Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).	Closed-loop recycling systems in manufacturing to recover materials.	Shifting to 100% recycled aluminium enclosures on many product lines.
Perceived Consumer	Reduce	Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.									
		All Apple offices, retail stores, and data centres run on 100% renewable energy.									
		Apple helps suppliers adopt renewable energy and reduce emissions.									
		Apple achieved zero waste to landfill at all established final assembly sites in 2024.									
	Reuse	Apple operates a global Trade-In and refurbishment program to give devices a second life.									
		Replacement and repair programs prolong device use (battery replacements, screen repairs).									
	Recycle	Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).									
		Closed-loop recycling systems in manufacturing to recover materials.									
		Shifting to 100% recycled aluminium enclosures on many product lines.									
Retail	Reduce	Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.									
		All Apple offices, retail stores, and data centres run on 100% renewable energy.									
		Apple helps suppliers adopt renewable energy and reduce emissions.									
		Apple achieved zero waste to landfill at all established final assembly sites in 2024.									
	Reuse	Apple operates a global Trade-In and refurbishment program to give devices a second life.									
		Replacement and repair programs prolong device use (battery replacements, screen repairs).									
	Recycle	Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).									
		Closed-loop recycling systems in manufacturing to recover materials.									
		Shifting to 100% recycled aluminium enclosures on many product lines.									

Table 4.6 – Reproduced Correlations

Consistent with the procedure outlined in RO3.1, factor scores will be computed for each respondent and used as inputs for subsequent group comparison analysis using one-way ANOVA.

7.6 Research Analysis 4.2 (for RA4)

Following factor analysis, factor scores representing the Reduce, Reuse, and Recycle dimensions will be computed for each respondent. One-way ANOVA will then be employed to compare mean factor scores across geographic regions.

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

$$H_1 : \text{At least one } \mu_k \neq \mu_l \text{ for } k \neq l$$

Where:

- μ_k denotes the mean sustainability factor score for geographic group k

Source of Variation	Sum of squares	Degree of Freedom	Mean Square	F Ratio	P-value
Between Group	SS_x	$c - 1$	$MS_x = \frac{SS_x}{c - 1}$	$\frac{MS_x}{MS_{error}}$	
Within Group	SS_{error}	$N - c$	$MS_{error} = \frac{SS_{error}}{N - c}$		
Total	SS_y	$N - 1$			

Table 4.7 - ANOVA

From Table 4.7, if the p-value is less than 0.05, the H_0 is rejected, suggesting significant differences in mean sustainability factor scores among the groups. Eta-squared (η^2) will be used to assess the magnitude of group differences, reflecting the proportion of variance in consumer perceptions. A larger eta-squared value suggests a stronger practical impact.

8. Future Research

Future research is recommended for RO3 and RO4, as both objectives reveal deeper behavioural and perceptual differences that warrant further investigation. For RO3, where generational differences in device retention drivers are identified, follow-up studies could examine how privacy expectations, ecosystem reliance, and technology usage evolve over time. Longitudinal research or comparative analysis with competitor ecosystems may further inform strategies to strengthen long-term customer loyalty.

Similarly, for RO4, where sustainability perceptions vary across geographic regions, future research could explore the cultural and regulatory factors shaping preferences for Reduce, Reuse, and Recycle initiatives. Region-specific qualitative studies may help Apple refine sustainability communication and implementation strategies to enhance global brand perception.

These objectives are prioritised for future research as privacy and ecosystem functions strongly influence device retention (Profwurzner, 2025), while sustainability perceptions play a key role in reinforcing long-term brand trust and loyalty (Blood, 2024).

9. Conclusion

9.1 Budget and Timescale

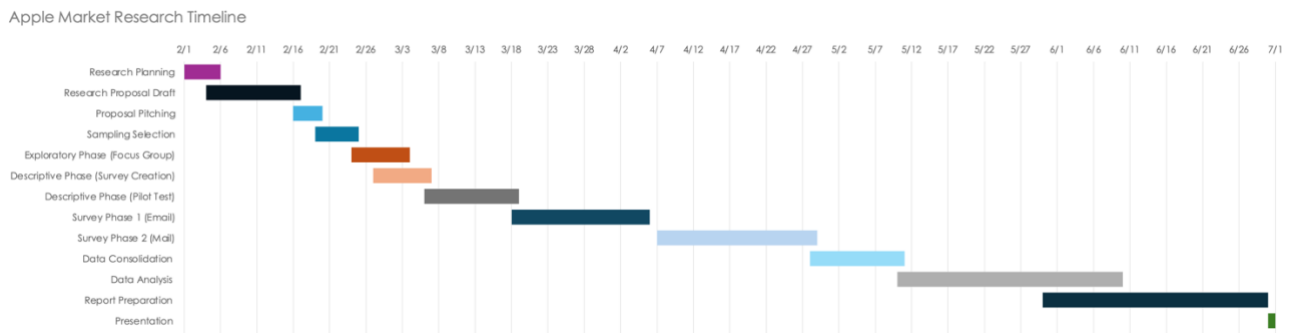


Figure 10.1 Gantt Chart

Having bi-weekly virtual meetings in addition to monthly physical meetings will support effective communication, assist progress updates, and allow us to hear feedback from your team.

Budget Allocation	
Focus Group	20%
Survey (Data Collection)	25%
Data Analysis	15%
Research Operation	10%
Incentives	15%
Administrative	5%
Other fees (Contingency, platform/ service charge)	5%
Additional Fund (For Sustainability donation if	5%
Total	100%

Figure 10.2 Budget Allocation Table

9.2 Summary

Through this research, Apple will obtain a more comprehensive understanding of consumer preferences for its products, services, ecosystem integration, and privacy features across major demographic and geographic segments. These insights will support more targeted product innovation, ecosystem development, and privacy communication strategies aligned with evolving user expectations.

The research will also examine how key economic factors—such as pricing sensitivity, income stability, and economic uncertainty—influence consumers’ purchase intentions. Findings will enable Apple to refine pricing, financing, and trade-in strategies to better respond to changing market conditions.

Alongside this, insights into consumer perceptions of Apple’s sustainability initiatives across the Reduce, Reuse, and Recycle dimensions will help identify opportunities to strengthen brand trust and reinforce long-term brand equity across regions.

We look forward to collaborating with Apple to deliver actionable, data-driven insights that support sustained growth, customer loyalty, strong privacy leadership, and responsible innovation.

10. Survey

Apple General Engagement Survey

Thank you for participating in this survey. This study explores consumers' preferences and experiences with Apple products and services.

Your responses are anonymous, confidential, and used solely for academic research. There are no right or wrong answers. The survey takes about 5–7 minutes, and your honest input is greatly appreciated.



Section 0 Qualifying Questions

Q1. Which of the following brands products do you currently own or regularly use?
(Select all that applies)

☐ Apple ☐ Microsoft ☐ Google ☐ Samsung ☐ Others

Q2. Which of the following Apple products or services do you currently own or regularly use?
(Select all that applies)

- ☐ iPhone
- ☐ Mac (Laptops: MacBook, as well as desktops: iMac®, Mac mini®, Mac Studio® and Mac Pro®)
- ☐ iPad (iPad Pro®, iPad Air®, iPad and iPad mini®)
- ☐ Wearables, Home and Accessories (Smartwatches, wireless headphones and spatial computers)
- ☐ Apple Services (e.g. iCloud, AppleCare, Apple TV+, etc.)

Section 1

Apple aims to expand its global presence by better understanding consumer preferences for its products and services. Your participation will provide valuable insights into how Apple can tailor its offerings to meet diverse consumer needs worldwide.

Q3. How important are the following factors when choosing an electronic device?
(1 = Not important at all, 7 = Extremely important)

	Not important at all				Extremely important			
	1	2	3	4	5	6	7	NA
Reliability and Brand Reputation								
Performance and Specifications								
Compatibility and Ecosystem								
Build Quality and Design								
Security and Privacy								

Q4. Overall, how satisfied are you with Apple's products and services?
(1 = Very dissatisfied, 7 = Very satisfied)

	Very dissatisfied				Very satisfied			
	1	2	3	4	5	6	7	
iPhone								
Mac (Laptops: MacBook, as well as desktops: iMac®, Mac mini®, Mac Studio® and Mac Pro®)								
iPad (iPad Pro®, iPad Air®, iPad and iPad mini®)								
Wearables, Home and Accessories (Smartwatches, wireless headphones and spatial computers)								
Apple Services (e.g. iCloud, AppleCare, Apple TV+, etc.)								

Q5. How important are the following Apple-specific product attributes when choosing your Apple products or services?
(1 = Not important at all, 7 = Extremely important)

	Not important at all				Extremely important			
	1	2	3	4	5	6	7	NA
Apple Silicon performance (A-series/M-series chips)								
ProMotion display (120Hz smoothness)								
Advanced camera features (Night mode, Photonic Engine, ProRAW)								
Dynamic Island (only for iPhone)								
Apple's cross-device ecosystem (AirDrop, Handoff, Universal Clipboard)								
Face ID and Secure Enclave								
App Tracking Transparency								
MagSafe wireless charging system								
Live Text and on-device intelligent features								
iCloud syncing and continuity features								

Remark session

For internal references

For Apple and Competitor Customers

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Section 2

Understanding how consumers respond to price changes is essential for developing effective pricing strategies. This survey explores willingness to pay and price sensitivity to support Apple's adaptive pricing decisions.

Q6. How likely are you to purchase an Apple product within the next 12 months?
(0 = Very unlikely, 100 = Very likely)

ANSWER: _____

Q7. Please indicate the extent to which you agree with the following statements.
(1 = Strongly disagree, 7 = Strongly agree)

	Strongly disagree										Strongly agree		
	1	2	3	4	5	6	7						
Rising prices make me think twice before buying new devices.													
Higher installments or financing costs make these devices feel less affordable to me.													
My current income and job stability influence whether I can afford to purchase an Apple device.													
When the economy feels uncertain, I avoid buying high-priced electronics.													

Section 3

Understanding how Apple's privacy protection practices and ecosystem integration shape users' perceptions and experiences. Your responses will contribute to assessing how privacy and seamless connectivity across devices influence customer trust and loyalty.

Q8. How likely are you to continue using Apple devices because of its ecosystem and privacy protection?
(0 = Very unlikely, 100 = Very likely)

ANSWER: _____

Q9. How important are the following factors about Apple's privacy and security features:
(1 = Strongly disagree, 7 = Strongly agree)

	Strongly disagree										Strongly agree		
	1	2	3	4	5	6	7						
Private Browsing: Safari keeps your activity private by not saving history, searches, or form data, blocking trackers, and locking windows when not in use.													
Password Monitoring: Checks to see whether your saved Keychain passwords have been compromised in data breaches.													
Location fuzzing: Apple Maps obscures your precise location by converting it to a less exact one after 24 hours and does not store your search or location history.													
Third-party app permissions: Apple's photo picker lets you control exactly which photos an app can access, shows what content apps can use, and provides reminders so you can change permissions.													
End-to-end encryption: iMessage and FaceTime messages are encrypted on your device so only you can access them, and Apple cannot read them in transit.													
In-store merchant transactions: Payments use a secure device number and dynamic code, keeping your card and personal info private unless you give permission.													

Section 4

This survey explores how Apple's sustainability initiatives influence consumer perceptions of the brand. Your feedback will provide insights into the impact of environmental responsibility on brand equity and purchase behaviour.

Q10. How will your positive feeling toward Apple be affected if you are aware that Apple has done the following?
(1 = Not at all affected, 7 = Extremely affected)

		Not at all affected									Extremely affected		
		1	2	3	4	5	6	7					
Reduce	Apple aims to make its entire supply chain and product life cycle carbon-neutral by 2030.												
	All Apple offices, retail stores, and data centres run on 100% renewable energy.												
	Apple helps suppliers adopt renewable energy and reduce emissions.												
	Apple achieved zero waste to landfill at all established final assembly sites in 2024.												
Reuse	Apple operates a global Trade-In and refurbishment program to give devices a second life.												
	Replacement and repair programs prolong device use (battery replacements, screen repairs).												
Recycle	Apple uses recycled materials across products (aluminium, rare earth elements, tin, cobalt).												
	Closed-loop recycling systems in manufacturing to recover materials.												
	Shifting to 100% recycled aluminium enclosures on many product lines.												

This is the end of the survey.

Your honest feedback is important in helping us improve our services. To maintain the integrity of this study, a limited number of participants may be randomly contacted to verify their responses. We would appreciate it if you could provide the following details.

Name: _____

Email: _____

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